

Self-Supervised Learning

Naeemullah Khan

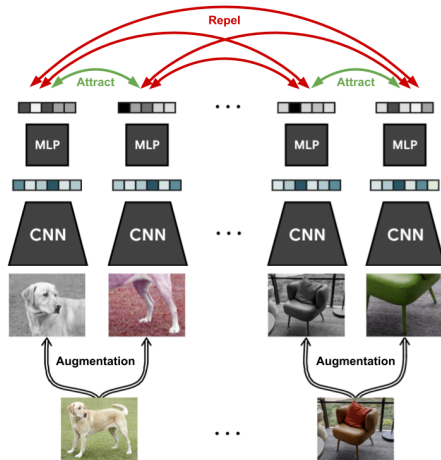
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Source: UvA Deep Learning Tutorials (P. Lippe), SimCLR tutorial; figure credit Chen et al. (SimCLR).

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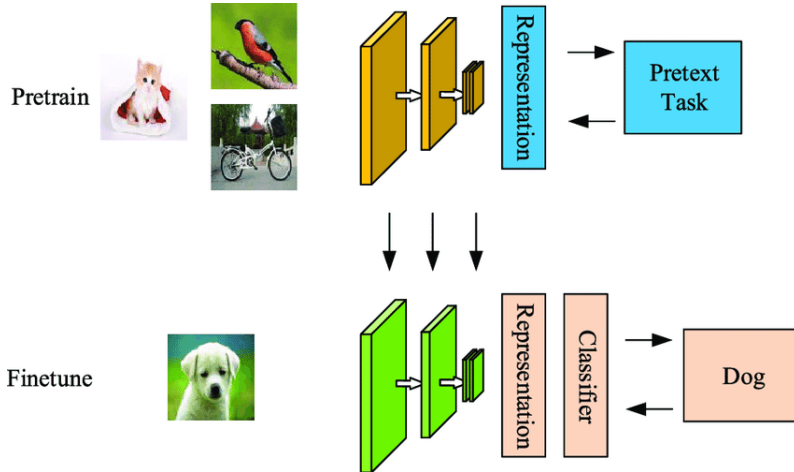
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5. A journey of methods:

The field has evolved from generative models (like VAEs and GANs), to contrastive and predictive coding, and now to hybrid approaches combining reconstruction and discrimination.



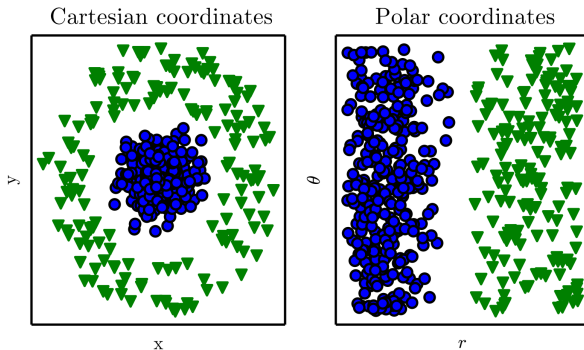
[Source: <https://framerusercontent.com/images/r0i6jmANAR5bsgzvqm0BJG9aas.png>]

By the end of this session, you should be able to:

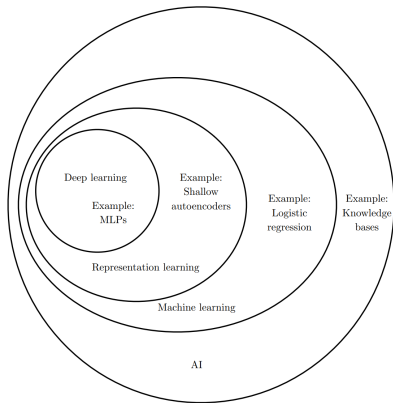
- ▶ **Explain** the data efficiency imperative and theoretical motivations behind self-supervised learning (SSL).
- ▶ **Identify** key cognitive principles that inspire self-supervision in machine learning.
- ▶ **Compare** reconstructive and contrastive pretext tasks used in SSL.
- ▶ **Formulate** the InfoNCE loss and explain its role in contrastive learning.
- ▶ **Describe** Instance Discrimination and **implement** the Momentum Contrast (MoCo) framework.

SSL: Introduction

Representations Matter



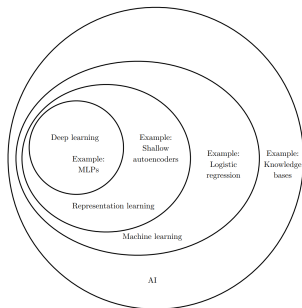
Source: Goodfellow, Bengio & Courville, *Deep Learning*, MIT Press, 2016, Fig. 1.1.



Deep Unsupervised Learning:

- ▶ Learns data representations without using labels.
- ▶ Is a subset of Deep Learning, which itself is a subset of Representation Learning, all within Machine Learning.

Source: Goodfellow, Bengio & Courville, *Deep Learning*, MIT Press, 2016, Fig. 1.4.



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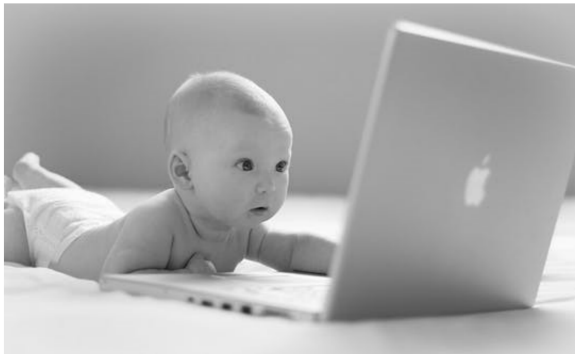
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Self-Supervised Learning (SSL):

- ▶ A form of unsupervised learning that creates supervision from the data itself by designing *pretext tasks*.
- ▶ These tasks generate pseudo-labels, enabling the model to learn useful representations without manual annotation.
- ▶ Often used interchangeably with unsupervised learning, but SSL specifically focuses on using the data's intrinsic structure.

- ▶ **High cost of dataset creation:** Each new task often requires building a new labeled dataset.
 - Involves preparing labeling manuals, defining categories, hiring annotators, building GUIs, and managing storage pipelines.
- ▶ **Expensive supervision:** High-quality labels can be costly or impractical to obtain (e.g., in medicine or legal domains).
- ▶ **Use unlabeled data:** The internet provides vast amounts of unlabeled images, videos, and text.
- ▶ **Cognitive motivation:** Animals and babies learn from their environment without explicit supervision, inspiring SSL approaches.



“Give a robot a label and you feed it for a second; teach a robot to label and you feed it for a lifetime.”

— Pierre Sermanet

This quote captures the core motivation behind self-supervised learning: machines that generate their own supervision from data reduce the reliance on costly manual labeling, and can keep learning at scale.

SSL is a type of unsupervised learning where the data itself provides supervision. Typically, part of the data is hidden, and a neural network is trained to predict it from the rest; this defines the *pretext task*. Well-chosen pretext tasks help models learn useful features without manual labels.

- ▶ **Pretext Task:** Examples include predicting masked patches, future representations (CPC), or distinguishing positive/negative pairs (contrastive).
- ▶ **Downstream Task:** After pre-training, the encoder is fine-tuned for tasks like classification or detection.
- ▶ **Key Trade-Offs:**
 - **Reconstructive** (e.g., autoencoders): capture low-level details, may miss semantics.
 - **Predictive/Contrastive** (e.g., CPC, MoCo): focus on high-level information.

What is Self-Supervised Learning? (cont.)

► “Pure” Reinforcement Learning (**cherry**)

- The machine predicts a scalar reward given once in a while.

► **A few bits for some samples**

► Supervised Learning (**icing**)

- The machine predicts a category or a few numbers for each input
- Predicting human-supplied data
- **10→10,000 bits per sample**

► Self-Supervised Learning (**cake génoise**)

- The machine predicts any part of its input for any observed part.
- Predicts future frames in videos
- **Millions of bits per sample**



Source: Y. LeCun, “Energy-Based Approaches to Representation Learning,” Institute for Advanced Study, 16 Oct 2019, slide 13.

SSL: Cognitive Principles

- ▶ **Predictive Coding:** The brain optimizes to predict future sensory inputs.
- ▶ **Error Correction:** Learning signals arise from reconstructive or contrastive errors.
 - *Reconstruction from Corrupted Inputs:*
 - ▶ Denoising autoencoders
 - ▶ Inpainting
 - ▶ Colorization, split-brain autoencoders
 - *Visual Common Sense Tasks:*
 - ▶ Relative patch prediction
 - ▶ Jigsaw puzzles
 - ▶ Rotation prediction
 - *Contrastive Learning:*
 - ▶ word2vec

- ▶ Contrastive Predictive Coding (CPC)
- ▶ Instance discrimination
- ▶ Recent state-of-the-art methods
- ▶ **Multi-View Perception:** Integrating different sensory views enhances representations.

SSL: Pretext Task Taxonomy

Pretext tasks are broadly categorized by their supervision signal and learning objective:

- ▶ **Reconstructive Methods:** Focus on reconstructing parts or properties of the input data.
- ▶ **Predictive Methods:** Involve predicting withheld or transformed aspects of the data.

Each category includes various techniques that have advanced SSL in computer vision.

Understanding this taxonomy helps in:

- ▶ Selecting suitable pretext tasks for specific applications.
- ▶ Gaining insight into how SSL methods learn transferable representations.

- ▶ **Corrupt input:** Introduce random noise to the input data, such as masking or adding Gaussian noise. The autoencoder is then trained to reconstruct the original, clean input from the corrupted version.
- ▶ **Robust feature learning:** By learning to recover the original data, the autoencoder is encouraged to extract meaningful and robust features that are less sensitive to noise and irrelevant variations.
- ▶ **Incremental understanding:** This approach helps the model generalize better by preventing it from simply memorizing the input, thus improving its ability to handle unseen or noisy data.

Denoising Autoencoders (cont.)

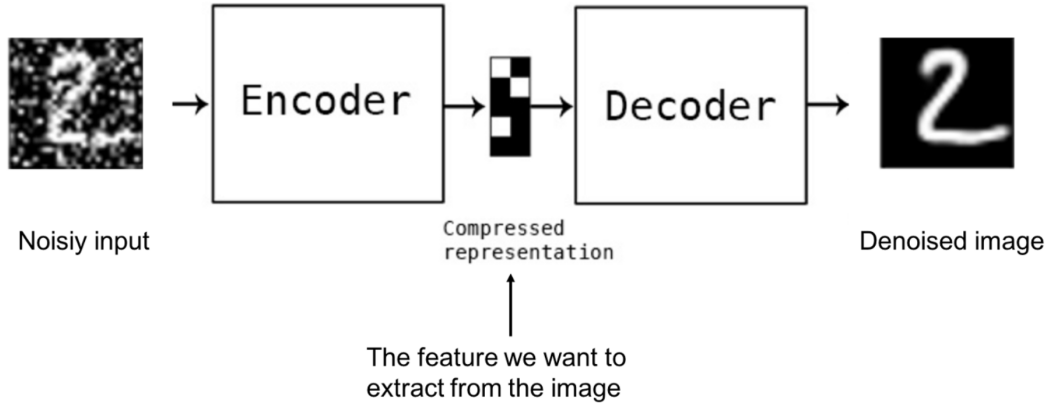
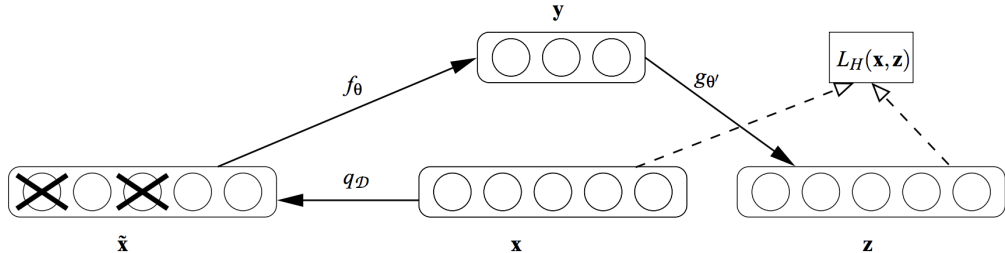


Figure source: neerajkrbansal1996, "Denoising-autoencoder.png," Wikimedia Commons, 2018 (CC0 1.0).



Vincent, Larochelle, Bengio & Manzagol (2008). Extracting and Composing Robust Features with Denoising Autoencoders. ICML 2008, Fig. 1.

- Additive isotropic *Gaussian noise* (GS): $\tilde{\mathbf{x}}|\mathbf{x} \sim \mathcal{N}(\mathbf{x}, \sigma^2 I)$;
- *Masking noise* (MN): a fraction v of the elements of $\mathbf{x}$ (chosen at random for each example) is forced to 0;
- *Salt-and-pepper noise* (SP): a fraction v of the elements of $\mathbf{x}$ (chosen at random for each example) is set to their minimum or maximum possible value (typically 0 or 1) according to a fair coin flip.

1

¹Vincent, Larochelle, Lajoie, Bengio & Manzagol (2010). Stacked Denoising Autoencoders. *JMLR* 11:3371-3408, Sec. 3.3–3.4.

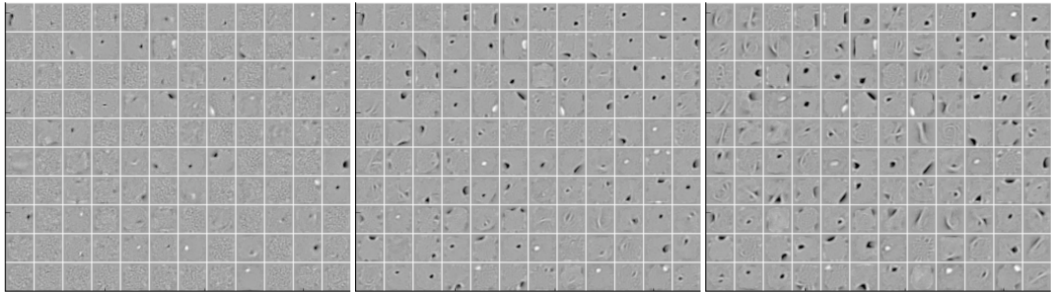
$$L_{2,\alpha}(\mathbf{x}, \mathbf{z}) = \alpha \left(\sum_{j \in \mathcal{J}(\tilde{\mathbf{x}})} (\mathbf{x}_j - \mathbf{z}_j)^2 \right) + \beta \left(\sum_{j \notin \mathcal{J}(\tilde{\mathbf{x}})} (\mathbf{x}_j - \mathbf{z}_j)^2 \right)$$

Vincent, Larochelle, Lajoie, Bengio & Manzagol (2010). Stacked Denoising Autoencoders. *JMLR* 11:3371-3408, Sec. 3.3–3.4.

$$L_{\text{IH},\alpha}(\mathbf{x}, \mathbf{z}) = \alpha \left(- \sum_{j \in \mathcal{J}(\tilde{\mathbf{x}})} [\mathbf{x}_j \log \mathbf{z}_j + (1 - \mathbf{x}_j) \log(1 - \mathbf{z}_j)] \right) \\ + \beta \left(- \sum_{j \notin \mathcal{J}(\tilde{\mathbf{x}})} [\mathbf{x}_j \log \mathbf{z}_j + (1 - \mathbf{x}_j) \log(1 - \mathbf{z}_j)] \right)$$

2

²Vincent, Larochelle, Lajoie, Bengio & Manzagol (2010). Stacked Denoising Autoencoders. *JMLR* 11:3371-3408, Sec. 3.3–3.4.



(a) No destroyed inputs

(b) 25% destruction

(c) 50% destruction

3

³Vincent, Larochelle, Bengio & Manzagol (2008). Extracting and Composing Robust Features with Denoising Autoencoders. ICML 2008, Fig. 3(a-c).

SSL: Instance Discrimination & Contrastive Learning

Pretext Task Evolution:

- ▶ Earlier: handcrafted tasks (e.g., jigsaw, colorization).
- ▶ Now: contrastive learning directly optimizes for invariance and discrimination.
- ▶ Leads to better, more transferable features and narrows the gap with supervised learning.

Instance Discrimination:

- ▶ Treats each image as its own class.
- ▶ Positive pairs: augmented views of the same image.
- ▶ Negative pairs: views from different images.
- ▶ Promotes invariance to augmentations and discrimination between instances.

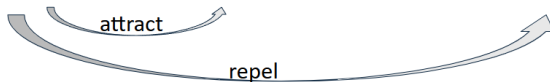
Contrastive Learning:

- ▶ Self-supervised approach for learning representations, especially in vision.
- ▶ Uses data structure instead of labels.
- ▶ Learns by pulling together similar (positive) pairs and pushing apart dissimilar (negative) pairs using a contrastive loss (e.g., InfoNCE).

Instance Discrimination is a self-supervised learning approach where each individual sample in the dataset is treated as a distinct class.

- ▶ The core idea is to learn representations by distinguishing between different instances, rather than relying on human-annotated labels.
- ▶ **Positive pairs** are generated by applying different augmentations (such as cropping, color jittering, or flipping) to the same image. These augmented views are considered to belong to the same class (i.e., the same instance).
- ▶ **Negative pairs** are formed by pairing augmented views of different images. These are treated as belonging to different classes (i.e., different instances).

Fundamentals of Instance Discrimination (cont.)



1. MoCo
2. SimCLR

Figure source: P. Abbeel, P. Chen, J. Ho, A. Srinivas, A. Li & W. Yan, "L7 Self-Supervised Learning," Deep Unsupervised Learning (CS294-158), UC Berkeley, Spring 2020.

Wu et al. (2018). Unsupervised Feature Learning via Non-Parametric Instance Discrimination. CVPR 2018.

- ▶ The learning objective is to **maximize agreement** (similarity) between representations of positive pairs, while minimizing agreement between negative pairs.
- ▶ This is typically achieved using a contrastive loss function, such as InfoNCE, which encourages the model to bring positive pairs closer in the embedding space and push negative pairs apart.
- ▶ Instance discrimination forms the basis for many popular self-supervised learning methods, such as MoCo, SimCLR, and PIRL.
- ▶ By maximizing agreement across views of the same instance, the model learns to extract meaningful and invariant features, which can be transferred to downstream tasks.

Momentum Contrast for Unsupervised Visual Representation Learning

Kaiming He Haoqi Fan Yuxin Wu Saining Xie Ross Girshick

Facebook AI Research (FAIR)

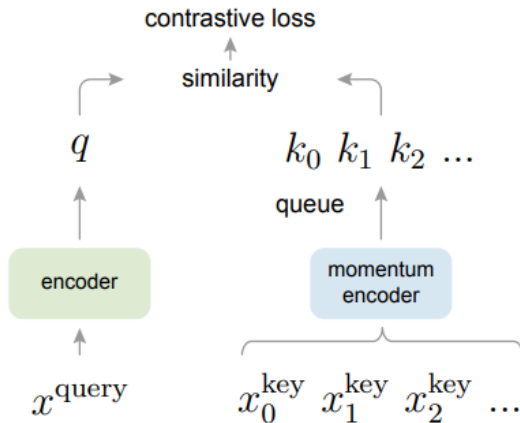
Momentum Contrast (MoCo) is a self-supervised learning framework designed for visual representation learning. Its key components are:

Query & Key Encoders: Two neural networks, f_q (query encoder) and f_k (key encoder), are used. The key encoder f_k is updated as an exponential moving average of the query encoder f_q :

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

where θ_k and θ_q are the parameters of f_k and f_q , and m is the momentum coefficient (e.g., $m = 0.999$).

Momentum Contrast (MoCo) (cont.)



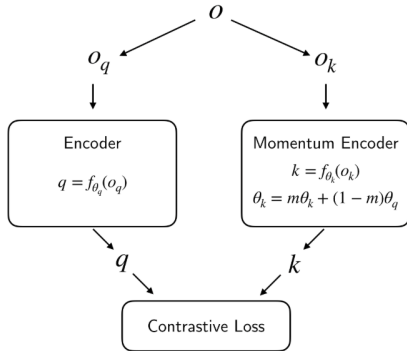
Source: He, Fan, Wu, Xie & Girshick, "Momentum Contrast for Unsupervised Visual Representation Learning," CVPR 2020, Fig. 1.

Dictionary Queue: MoCo maintains a large queue (dictionary) of encoded keys from previous batches. This enables the use of a large and consistent set of negative samples for contrastive learning, which is crucial for effective representation learning.

Contrastive Loss: The InfoNCE loss is used to train the encoders. For a given query q and its positive key k^+ , along with a set of negative keys $\{k_0, k_1, \dots, k_N\}$ from the dictionary, the loss is:

$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{i=0}^N \exp(q \cdot k_i / \tau)}$$

where τ is a temperature hyperparameter.



$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k_+ / \tau)}{\sum_{i=0}^K \exp(q \cdot k_i / \tau)}$$

Left figure: Laskin, Srinivas & Abbeel, "CURL: Contrastive Unsupervised Representations for Reinforcement Learning," ICML 2020, Fig. 1 (adapted).

Source: He, Fan, Wu, Xie & Girshick, "Momentum Contrast for Unsupervised Visual Representation Learning," CVPR 2020, Eqn. (1).

Key Steps in MoCo Training:

1. For each image, generate two augmentations: one for the query encoder (f_q), one for the key encoder (f_k).
2. Encode the query and key.
3. Compute the InfoNCE loss using the current positive key and the dictionary of negative keys.
4. Update f_q via backpropagation; update f_k via momentum.
5. Enqueue the new key and dequeue the oldest key to maintain the dictionary size.

Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

```
# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
    x_q = aug(x) # a randomly augmented version
    x_k = aug(x) # another randomly augmented version

    q = f_q.forward(x_q) # queries: NxK
    k = f_k.forward(x_k) # keys: NxK
    k = k.detach() # no gradient to keys

    # positive logits: Nx1
    l_pos = bmm(q.view(N,1,K), k.view(N,K,1))

    # negative logits: NxK
    l_neg = mm(q.view(N,K), queue.view(K,K))

    # logits: Nx(1+K)
    logits = cat([l_pos, l_neg], dim=1)

    # contrastive loss, Eqn.(1)
    labels = zeros(N) # positives are the 0-th
    loss = CrossEntropyLoss(logits/t, labels)

    # SGD update: query network
    loss.backward()
    update(f_q.params)

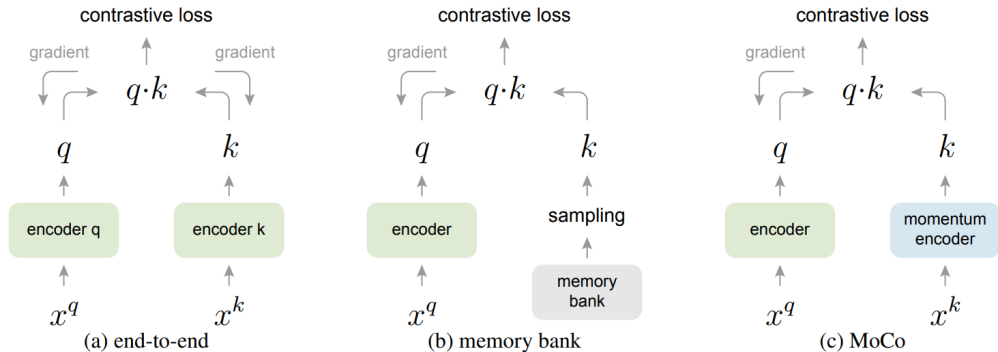
    # momentum update: key network
    f_k.params = m*f_k.params+(1-m)*f_q.params

    # update dictionary
    enqueue(queue, k) # enqueue the current minibatch
    dequeue(queue) # dequeue the earliest minibatch
```

bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.

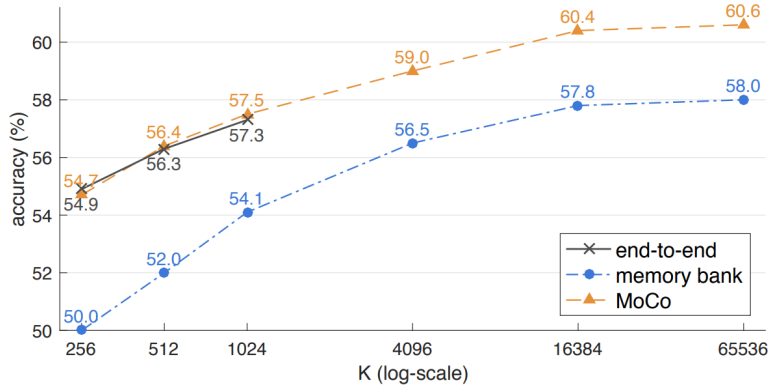
Source: He, Fan, Wu, Xie & Girshick, “Momentum Contrast for Unsupervised Visual Representation Learning,” CVPR 2020, Algorithm 1.

Momentum Contrast (MoCo) (cont.)



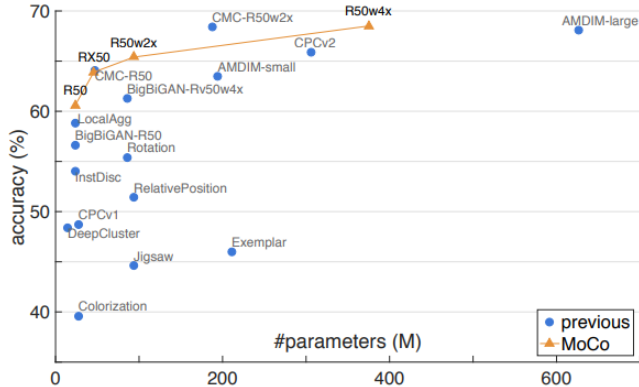
Source: He, Fan, Wu, Xie & Girshick, "Momentum Contrast for Unsupervised Visual Representation Learning," CVPR 2020, Fig. 2.

Momentum Contrast (MoCo) (cont.)



Source: He, Fan, Wu, Xie & Girshick, "Momentum Contrast for Unsupervised Visual Representation Learning," CVPR 2020, Fig. 3.

Momentum Contrast (MoCo) (cont.)



Source: He, Fan, Wu, Xie & Girshick, "Momentum Contrast for Unsupervised Visual Representation Learning," CVPR 2020, Table 1 (figure).

Key Advantages:

- ▶ Enables large and consistent negative sets for contrastive learning.
- ▶ Momentum update stabilizes the key encoder, improving training.
- ▶ Scalable to large datasets and high-dimensional representations.

SSL: References

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